QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 13

RULE · Ratio (nisbat): part-to-part vs part-to-whole

A $2 : 3$ ratio has $2 + 3 = 5$ total parts. Each share $= \dfrac{\text{its number}}{\text{total parts}}$.

20 total, ratio $2 : 3 \implies \dfrac{2}{5}(20) = 8$ and $\dfrac{3}{5}(20) = 12$

RULE · Unit rate

$$\text{unit rate} = \frac{\text{quantity}}{\text{time (or other unit)}}$$

180 km in 3 h $\implies 180 \div 3 = 60$ km/h

RULE · Solving a proportion (proportsiya)

$$\frac{a}{b} = \frac{c}{d} \implies a \times d = b \times c$$

$$\frac{x}{5} = \frac{12}{20} \implies 20x = 60 \implies x = 3$$

Flour:sugar $= 3 : 5$, 18 cups flour $\implies \dfrac{3}{5} = \dfrac{18}{s} \implies s = 30$

Keep the same quantity on top on both sides, and match units before you set up the proportion.